THE RELATIVE DIAGONAL IS AN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4263-4276, following the statement at 4254-4261.

Allowed standard prerequisites (editorial).

- Fibre products of affines and open fibre products. lemma-fibre-product-affines and lemma-open-fibre-product (same chapter, cited at ~4266-4267). V x_U V is affine open in X x_S X.
- Local target criterion for closed immersions. lemma-closed-local-target (same chapter, cited at ~4270). Reduces to checking each piece V x_U V.
- Affine diagonal is a closed immersion. lemma-diagonal-affines-closed (same chapter, cited at ~4275). Concluding engine on each piece.

Source reference. Original lemma 01HL: sources/stacks-schemes/pinned-original-source.tex, source lines 574–574. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KI: sources/stacks-schemes/pinned-original-source.tex, source lines 4239–4239. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JQ: sources/stacks-schemes/pinned-original-source.tex, source lines 3399–3399. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JR: sources/stacks-schemes/pinned-original-source.tex, source lines 3415–3415. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme over S. The diagonal morphism $\Delta_{X/S}$ is an immersion.*

Proof. Recall that if $V \subset X$ is affine open and maps into $U \subset S$ affine open, then $V \times_U V$ is affine open in $X \times_S X$, see Lemmas 01JQ and 01JR. Consider the open

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subscheme W of $X \times_S X$ which is the union of these affine opens $V \times_U V$. By Lemma 01HL it is enough to show that each morphism $\Delta_{X/S}^{-1}(V \times_U V) \to V \times_U V$ is a closed immersion. Since $V = \Delta_{X/S}^{-1}(V \times_U V)$ we are just checking that $\Delta_{V/U}$ is a closed immersion, which is Lemma 01KI. $\square$